

We saw at the end of the last discussion that if we consider the real-valued bivariate function $z = f(x, y)$ and restrict our view to a specific level curve, $f(x, y) = c$, then the gradient at any point on this curve is normal to the curve itself:

Thus, if we extend to a real-valued trivariate function $w = f(x, y, z)$ and restrict our view to a specific level *surface*, $f(x, y, z) = c$, then the gradient at any point on this surface is normal to the surface itself. This allows us to use the gradient to define the tangent *plane*.

Definition. The **tangent plane** at the point $P_0(x_0, y_0, z_0)$ on the level surface $f(x, y, z) = c$ of a differentiable function f is the plane through P_0 normal to $\nabla f(x_0, y_0, z_0)$. The **normal line** of the surface at P_0 is the line through P_0 parallel to $\nabla f(x_0, y_0, z_0)$.

Example. Find the tangent plane and normal line of the level surface

$$f(x, y, z) = x^2 + y^2 + z - 9 = 0$$

at the point $P_0(1, 2, 4)$

Everything we've done so far has been with respect to a given real-valued trivariate function $w = f(x, y, z)$ and then looking at its level surfaces.

However, what if we *start* with a given surface? Clearly tangent planes still exist, so how do we compute them using the gradient? Well, observe that a surface given by the graph of $f(x, y)$ is the zero-level surface of the trivariate function

$$F(x, y, z) = f(x, y) - z.$$

Then, tangent planes to f 's surface become tangent planes to F 's level surface at zero. Thus, F 's gradient is given by

$$\nabla F(x, y, z) = \langle f_x(x, y), f_y(x, y), -1 \rangle$$

so a surface at some point $P(x_0, y_0, f(x_0, y_0))$ is given by

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - z_0) = 0.$$

Example. Find the plane tangent to the surface $z = x \cos(y) - ye^x$ at $P(0, 0, 0)$.

In single-variable calculus, we consider functions like $y = f(x)$. If we localize to a single value, say, x_0 , and 'zoom in', we find that the tangent line $y - y_0 = f'(x_0)(x - x_0)$ is a decent approximation of the function:

We can rewrite this tangent line equation as

$$y = y_0 + f'(x_0)(x - x_0).$$

Such an object is called a *linearization* of the function $f(x)$.

In multiple dimensions, we consider the real valued bivariate function $z = f(x, y)$ and yield

$$L(x, y) = f(x_0, y_0) + \nabla f(x_0, y_0) \cdot \langle x - x_0, y - y_0 \rangle$$

or, equivalently,

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

This is the multi-dimensional linearization of f and, since $f(x, y) \approx L(x, y)$, this is sometimes called the **standard linear approximation** of f at (x_0, y_0) . Note that this is a *plane*! This is sometimes called the *tangent-plane approximation*.

We can extend another notion from single-variable calculus: the differential. In single-variable calculus, we can think of $dx = \Delta x = x - x_0$ and ‘write’ the following:

$$\frac{df}{dx} = f'(x_0) \iff df = f'(x_0)dx$$

In thinking of it this way, we get the following definition:

Definition. Given a single-variable function $y = f(x)$, the **differential** of f at x_0 is given by

$$df = f'(x_0)dx$$

Roughly speaking, the differential tells us what the net-change of the function values is when we change allow x to vary near x_0 .

For a multi-variate function $z = f(x, y)$, we arrive at the following:

Definition. Given a real-valued bivariate function $z = f(x, y)$ and a point $P(x_0, y_0)$, if we move to the point $Q(x + x_0, y + y_0)$, the resulting change

$$\begin{aligned} df &= \nabla f(x_0, y_0) \cdot \langle dx, dy \rangle \\ &= f_x(x_0, y_0)dx + f_y(x_0, y_0)dy \end{aligned}$$

is called the **total differential** of f .